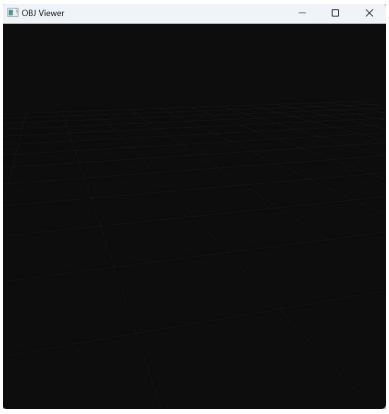
Project-2

Finish

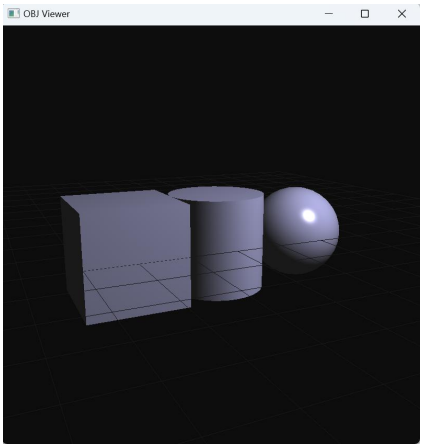
A. Camera Controls + Grid



B. i. Open an obj file

```python
def drop_callback(window, paths):
    global models
    offset = len(models) * 2.0
    for path in paths:
        try:
            model = ObjModel(path, offset)
            models.append(model)
        except Exception as e:
            print(f"Failed to load {path}: {e}")
```

ii. Read the obj file and display the mesh

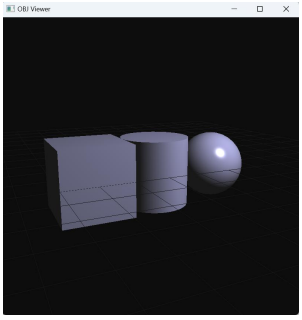


iii. Print out the following information

```
PS D:\大学课程\25-1\计算机图形学\project-2\Project2-main> & D:/anaconda3/envs/cg_lab/pyt

[Loaded] D:\大学课程\25-1\计算机图形学\project-2\Project2-sample-objs\cube-tri.obj
Total faces: 12
Triangles: 12
Quads:     0
Ngons:     0

[Loaded] D:\大学课程\25-1\计算机图形学\project-2\Project2-sample-objs\cylinder-tri.obj
Total faces: 124
Triangles: 124
Quads:     0
Ngons:     0

[Loaded] D:\大学课程\25-1\计算机图形学\project-2\Project2-sample-objs\sphere-tri.obj
Total faces: 960
Triangles: 960
Quads:     0
Ngons:     0
```

iv. Support for Opening Multiple OBJ Files:

```python
def drop_callback(window, paths):
    global models
    offset = len(models) * 2.0
    for path in paths:
        try:
            model = ObjModel(path, offset)
            models.append(model)
        except Exception as e:
            print(f"Failed to load {path}: {e}")
```



v. Support for Meshes with Polygons of Varying Vertex Counts:

```
[Loaded] D:\大学课程\25-1\计算机图形学\project-2\Project2-sample-objs\cube-tri-quad.obj
Total faces: 10
Triangles: 8
Quads:     2
Ngons:     0

[Loaded] D:\大学课程\25-1\计算机图形学\project-2\Project2-sample-objs\cylinder-tri-quad-n.obj
Total faces: 45
Triangles: 22
Quads:     21
Ngons:     2

[Loaded] D:\大学课程\25-1\计算机图形学\project-2\Project2-sample-objs\sphere-tri-quad.obj
Total faces: 524
Triangles: 88
Quads:     436
Ngons:     0
```

C. Lighting & Etc

In this project, I implemented Phong Illumination and Phong Shading to render all loaded .obj models. The fragment shader computes ambient, diffuse, and specular components per fragment using the classic Phong lighting model